Persuasion and Receiver's Risk Attitudes *

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Abstract

This paper studies how the receiver's risk attitudes affect the set of implementable outcomes in a persuasion problem. I consider a two-period consumption model, where the receiver's future income is uncertain and he has the option to transfer his wealth intertemporally through saving. I present a local comparative statics result, showing that any marginal change in the receiver's prudence either expands or shrinks the set of implementable persuasion outcomes. Risk aversion does not affect persuasion outcomes, except through pivoting the receiver's optimal action under the prior belief. Other higher order risk attitudes have no direct impact on persuasion outcomes.

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1 Introduction

In a Bayesian persuasion problem, an information sender (she) designs an experiment for an information receiver (him), in order to persuade the receiver to change his action. The literature of persuasion and information design is pioneered by Kamenica and Gentzkow (2011) and Rayo and Segal (2010). [1]. Since a persuasion problem is all about manipulating the receiver's posterior beliefs to pivot his action under uncertainty, it is natural to ask how the receiver's [2] risk attitudes is going to affect the persuasion outcomes.

The existing literature often explicitly or implicitly impose restrictions on the receiver's risk attitudes. For example, many studies assume the receiver's optimal action only depends on the posterior mean of the uncertain state. Gentzkow and Kamenica (2016) provides a general characterization of the solutions for the class of persuasion problems where the receiver's optimal action depends on the posterior mean. Dworczak and Martini (2019) explicitly assumes the sender's utility depends on the mean of posterior beliefs to establish the equivalence between a persuasion problem and a consumer choice problem with a price scheme. Kolotilin (2018) and Kolotilin, Mylovanov, Zapechelnyuk, and Li (2017) study persuasion problems where the receiver has private information. They focus on binary action spaces and assume the receiver's utility to be linear in the state space if the receiver chooses to act. The linearity assumption implies that the receiver's optimal action only depends on the posterior mean. Assuming the receiver only responds to changes in the posterior mean might be too restrictive. In many applications, the sender chooses the informativeness of the experiment by adding or removing white noises. But adding or removing white noises preserves the mean and thus has no effect on these persuasion problems.

This paper studies how persuasion outcomes are affected by the receiver's risk attitudes. I consider a simple two period consumption problem, where the receiver faces uncertainty in his future income. The receiver's action space is assumed to be binary, which contains the action of consuming and the action of saving. The receiver has the ability to transfer his wealth intertemporally through saving. The sender persuades the receiver to change his actions by choosing an experiment about the receiver's income in the second period. I show the receiver's prudence plays a key role in determining the set of implementable persuasion outcomes. The receiver's risk aversion also affects persuasion outcomes, but only through the receiver's optimal action under the prior belief. Any other higher order risk attitude has

[1] See Ostrovsky and Schwarz (2010) and Calzolari and Pavan (2006) for other early works in this literature and Bergemann and Morris (2019) and Kamenica (2019) for an excellent literature reviews

[2] It is also interesting to understand how the sender's risk attitudes affect persuasion outcomes. A recent study by Kolotilin, Mylovanov, and Zapechelnyuk (2019) characterizes the class of sender's utility functions, for which an upper censorship is optimal.

no direct impact on persuasion outcomes.

The main finding of this paper, Theorem 1, is a local comparative statics result. I show if the receiver becomes more prudent and this change is marginal, then the set of implementable persuasion outcomes is either shrinking or growing, depending on the receiver's optimal action under the prior belief. In particular, the set of implementable persuasion outcomes becomes larger if the receiver prefers consuming under the prior belief. The set changes in the opposite direction if the receiver prefers saving under the prior belief. The intuition for this result is aligned with the incentives of precautionary savings in Kimball (1990), where the concept of prudence was initially introduced. Kimball (1990) showed a more prudent agent tends to have more precautionary savings to smooth intertemporal consumption streams. Since it is optimal for the receiver to consume under the prior belief, it must be easier for any experiment to persuade him to save if he becomes more prudent. Theorem 1 is a local comparative statics result in the sense that its conclusion fails if the change in receiver's risk attitudes is large enough to pivot the receiver's optimal action under the prior belief. One implication of this observation is risk aversion can only affect persuasion outcomes through the receiver's optimal action under the prior belief.

The intertemporal consumption model considered in this paper, which is similar with the one in Kimball (1990), might seem too specific. It turns out this consumption model has various alternative interpretations. Eeckhoudt and Schlesinger (2006) presents an alternative definition of prudence and this definition allows the idea of prudence to be extended beyond any specific consumption model. For example, the two periods in the consumption model can be reinterpreted as two mutually exclusive random events, and there is an additional source of uncertainty in one the events. The receiver can then transfer his assets between these two events through insurances or investments.

This paper contributes broadly to both the literature of risk attitudes and the literature of persuasion. To my knowledge, this is the first paper that explicitly studies the connections between receiver's risk attitudes and the set of implementable persuasion outcomes.

2 Model

In a standard Bayesian persuasion problem, an information sender designs an experiment about an uncertain state of the world to persuade a Bayesian updating receiver.

2.1 Receiver's problem

In this section, I consider a two-period consumption model. There are two periods $t \in \{0, 1\}$ and the receiver's lifetime utility depends on consumption in both periods. The receiver's income at $t = 0$ is normalized to be zero and his income at $t = 1$ is denoted as ω. The receiver consumes $c \in \mathbb{R}$ at $t = 0$. At $t = 1$, the receiver consumes the remaining of his wealth after ω is realized and negative consumption is allowed. It follows that the receiver's lifetime utility function U can be defined as a function of c. U is further assumed to be additively separable across the two periods.[3]

$$U(c; \omega) := u(c) + u(\omega - c)$$

The receiver's risk attitudes are determined by the shape of the utility function u. The utility function u is assumed to be smooth, strictly increasing $u' > 0$ and strictly concave $u'' < 0$ (risk averse) on some closed intervals. The set of utility functions that satisfies these assumptions is denoted as $\mathcal{U}$.

The receiver's income ω at $t = 1$ is stochastic. There is a public prior belief π_0 over the state of the world $\omega \in \Omega := [0, 1]$. Under uncertainty, the receiver's lifetime utility is assumed to take an expected utility form. With a generic belief π, the receiver's lifetime expected utility of consuming c at $t = 0$ is given by

$$E_\pi[U(c)] = u(c) + E_\pi[u(\omega - c)]$$

Suppose the receiver knows the realization of ω at $t = 0$. It is easily seen the receiver maximizes the lifetime utility by consuming $\frac{\omega}{2}$ in both periods. So $\frac{1}{2}, 0$ are interpreted as the most optimistic/pessimistic consumption level at $t = 0$. To simplify the persuasion problem, the receiver's action space A is restricted to be binary, $A := \{\overline{c}, \underline{c}\}$ with $0 \leq \underline{c} < \overline{c} \leq \frac{1}{2}$. The action $\underline{c}$ is called "saving" and the action $\overline{c}$ is called "consuming".

The receiver's problem is to choose an action from A to maximize his lifetime expected utility. Intuitively, the receiver chooses consuming if his future income ω is sufficiently large to be above a certain threshold. It turns out with the binary action space, changes in receiver's risk attitudes have no effects on his preferences under certainty. That is, the threshold is independent of the shape of the utility function u.

Proposition 1. *Assume certainty, that is, the realization of ω is assumed to be known at $t = 0$. Let $u_1, u_2 \in \mathcal{U}$ and $\succeq_1, \succeq_2$ be the corresponding preferences over consumption streams.*

[3] U can be interpreted as the most commonly used exponential discounting form, and the discounting factor is assumed to be 1 for simplicity.

Then $(\overline{c}, \omega - \overline{c}) \succeq_1 (\underline{c}, \omega - \underline{c})$ *if and only if* $(\overline{c}, \omega - \overline{c}) \succeq_2 (\underline{c}, \omega - \underline{c})$ *if and only if* $\omega \geq \overline{c} + \underline{c}$.

The goal of this paper is to examine the impacts of receiver's risk attitudes on persuasion outcomes. Proposition 1 implies that the receiver's preferred action under certainty is unaffected by his risk attitudes as long as he is risk averse. It follows his preference under certainty can be risk attitudes can affect persuasion outcomes only through the receiver's preferences under uncertainty. One difficulty of extending the model in this paper to accommodate for arbitrary finite action spaces is that a result like Proposition 1 cannot be obtained in general when there are more than two actions. Without an analogue to Proposition 1, it will not be clear whether the persuasion outcomes are affected by the receiver's preference under certainty or his risk attitudes.

The specific consumption model considered in this paper is similar to the model used by Kimball (1990) to motivate the connection between precautionary saving and prudence. It turns out this model can be interpreted in various alternative ways such as insurance and investment problems. For example, consider an agent who faces two mutually exclusive random events which are equally likely to occur. The agent is allowed to transfer his wealth (normalized to zero) between these two events by purchasing costless insurances. Say his wealth in the first event is c so his wealth in the second event is $-c$. Suppose in the second event, there is an additional risk in the form of a random shock ω to his wealth. One can easily see the agent's insurance problem has exactly the same form as the consumption problem. Eeckhoudt and Schlesinger (2006) showed this agent is willing to buy insurance for the second event, that is, he prefers to have the additional risk when he is richer, if and only if he is prudent. In the context of persuasion, the insurance interpretation has many applications. One example can be that the receiver underestimates the likelihood of the first event, while a benevolent information sender has the correct assessment of the likelihoods. One can immediately see this persuasion problem with the insurance interpretation is equivalent to the one with the precautionary saving interpretation when the sender has a higher discount factor than the receiver.

2.2 Sender's problem

The sender's utility depends on the receiver's action $c \in A$ and the state of the world $\omega \in \Omega$. Her utility function is a generic function $V : A \times \Omega \to \mathbb{R}$ that is continuous on Ω and the sender is also assumed to be an expected utility maximizer. The sender can commit to an experiment (or a signal structure) to persuade the receiver. An experiment S is defined as a measurable mapping from the state space to the set of probability distributions over a signal space Θ, that is, $S : \Omega \to \triangle\Theta$.[4]

It is without loss of generality to assume $\Theta = A = \{\overline{c}, \underline{c}\}$, that is, the signals are direct. To see this, let $\hat{\Theta}$ be a generic signal space. After receiving a signal $\theta \in \hat{\Theta}$, the receiver updates his belief and chooses the optimal action c_θ from A to maximize his expected utility based on his posterior belief.[5] Therefore, elements of $\{\theta \in \hat{\Theta} : c_\theta = \underline{c}\}$ are relabelled as $\underline{c}$ since the receiver's responses to these signals are the same. The rest of the signals are similarly relabelled as $\underline{c}$. The direct signals can be interpreted as recommendations of actions.

Denote $p(\omega)$ as the probability of recommending saving, conditional on the true state being $\omega \in \Omega$. Note that an experiment is completely determined by $p : \Omega \to [0, 1]$ when the action space is binary. The sender's problem is to choose an optimal experiment, or equivalently, the function p, to maximize her expected utility.

Let $\pi_{\underline{c}}, \pi_{\overline{c}}$ be the receiver's posterior beliefs after receiving the direct signals $\underline{c}, \overline{c}$, respectively. In order to convince the receiver that it is indeed optimal to follow the recommendations, the experiment has to satisfy the following *obedience* constraints.

$$E_{\pi_{\underline{c}}}[U(\underline{c})] \geq E_{\pi_{\underline{c}}}[U(\overline{c})] \tag{1}$$

$$E_{\pi_{\overline{c}}}[U(\overline{c})] \geq E_{\pi_{\overline{c}}}[U(\underline{c})] \tag{2}$$

An experiment p is said to be **implementable** if it satisfies the obedience constraints. It turns out one of the above two obedience constraints must be redundant if the other one is satisfied, depending on the prior belief π_0.[6]

Lemma 1. *If $\overline{c}$ is optimal under the prior π_0 and the obedience constraint (1) is satisfied, then the constraint (2) is redundant; if $\underline{c}$ is optimal under the prior π_0 and the obedience constraint (2) is satisfied, then the constraint (1) is redundant.*

The intuition behind Lemma 1 is as follows: it is optimal for the receiver to choose consuming at $t = 0$ if he is sufficiently optimistic about his future income ω, according to the prior belief π_0. To pivot his decision from consuming to saving in (1), his updated posterior belief $\pi_{\underline{c}}$, after receiving the signal of saving, must be more pessimistic than the prior belief. In Bayesian updating, the expectation of the posterior beliefs is equal to the prior belief (see Kamenica and Gentzkow (2011)). Thus his posterior belief $\pi_{\overline{c}}$, after receiving the signal of consuming, must be more optimistic than the prior belief. It follows the obedience constraint (2) is automatically satisfied.

[4]$\triangle\Theta$ denotes the set of all probability distributions over Θ.

[5]The argument requires a tie-breaking rule to be assumed. For simplicity, it is assumed the receiver breaks the tie in the favor of the sender when he is indifferent between the two actions.

[6]See Bergemann and Morris (2016) for a similar result in a more general setting.

3 Comparative Statics

This section explores the impacts of the receiver's risk attitudes on persuasion outcomes. I consider two receivers with different risk attitudes, which are characterized by their utility functions u_1, u_2. I adopt the Arrow-Pratt measure of absolute risk aversion (Arrow (1965) and Pratt (1964)) and the absolute measure of prudence proposed by Kimball (1990).

Definition 1. (Comparative Risk Aversion and Prudence)
For $u_1, u_2 \in \mathcal{U}$, u_1 is said to be more risk averse than u_2 if $-\frac{u_1''(c)}{u_1'(c)} > -\frac{u_2''(\hat{c})}{u_2'(\hat{c})}$ for all $c, \hat{c}$; u_1 is said to be more prudent than u_2 if $-\frac{u_1'''(c)}{u_1''(c)} > -\frac{u_2'''(\hat{c})}{u_2''(\hat{c})}$ for all $c, \hat{c}$.

The comparative measures of risk aversion and prudence are stronger than standard ones, which the inequalities are only required to hold point-wise. The following result establishes a connection between the prudence and the set of implementable experiments.

Theorem 1. *Let $u_1, u_2 \in \mathcal{U}$, and u_1 be more prudent than u_2.*

1. *If consuming is optimal for both u_1, u_2 under the prior belief, then any implementable experiment for u_2 is also implementable for u_1.*

2. *If saving is optimal for both u_1, u_2 under the prior belief, then any implementable experiment for u_1 is also implementable for u_2.*

Theorem 1 says if u_1 is more prudent than u_2 and their optimal actions under the prior belief are the same, then the set of persuasion outcomes for u_1 is either a subset or super set of u_2. The intuition for Theorem 1 is related to the incentives of precautionary savings by prudence Kimball (1990). Taking the first part of Theorem 1 as an example. Kimball (1990) showed consumers tend to have more precautionary savings if they are more prudent. Since it is optimal for both u_1, u_2 to consume under the prior belief, it is easier for any experiment to persuade u_1 to save since u_1 is more prudent.

Theorem 1 covers two cases when the optimal actions for u_1, u_2 are the same under the prior belief but these two cases are not exhaustive. The following proposition discusses the remaining two possible cases.

Proposition 2. *Let $u_1, u_2 \in \mathcal{U}$, and u_1 be more prudent than u_2.*

1. *If consuming is optimal for u_1 under the prior belief, then consuming is also optimal for u_2 under the prior belief.*

2. *If saving is optimal for u_1 under the prior belief and consuming is optimal for u_2 under the prior belief, then there exists an experiment implementable for u_1 but not for u_2; and there exists an experiment implementable for u_2 but not for u_1.*

In view of Proposition 2, Theorem 1 can be interpreted as a *local* comparative statics result. Suppose the magnitude of the change in the receiver's prudence is only marginal such that his optimal action under the prior belief is unchanged. Then the direction of changes in the set of persuasion outcomes is completely determined by his optimal action under the prior belief. In many applications, the receiver is often assumed to have a utility function with constant absolute risk aversion, which also has constant absolute prudence. Researchers are typically interested in comparative statics results with respect to a marginal change in the risk aversion (prudence) parameter and Theorem 1 is readily useful.

Another takeaway from Proposition 2 and Theorem 1 is that the receiver's risk aversion influences the persuasion outcomes only through the receiver's optimal choice under the prior belief, since the set of implementable experiments is unaffected by any change in receiver's risk aversion as long as his prudence is unchanged. Moreover, other higher order risk attitudes have no impact on the persuasion outcomes.

3.1 Application

The main comparative statics result does not depend on the sender's utility function V, as long as the sender's problem has a solution. In this section, I present a model of consumption boost by the government and show a direct application of the comparative statics result. In the model, the sender is a benevolent government and the receiver is a representative consumer.

The government wants to increase consumer spending at $t = 0$ to help economic recovery after the pandemic of COVID. This can be done by a variety of policies such as a direct stimulus paycheck. The model in this section focuses exclusively on policies that boost consumer confidence through the information channels. For simplicity, consider the following utility function V for the government.

$$V(c) =: u(c)$$

The government's objective is to maximize the expected consumption at $t = 0$ regardless of the state of the world. With the binary action space, the government chooses an experiment to persuade the consumer to increase the probability of consuming at $t = 0$. To explore the impact of a change in consumer's prudence on the government's payoff, suppose the consumer's utility function changes from u_1 to u_2 and u_1 is more prudent.

If the optimal action is consuming for both u_1, u_2 under the prior belief, then the government can choose a non-informative experiment so both u_1, u_2 choose consuming with probability one. If saving is optimal for both u_1, u_2 under the prior belief, then by Theorem 1, any implementable experiment for u_1 is also implementable for u_1. It follows the government can achieve a higher payoff with u_2 than with u_1. By Proposition 2, it is impossible that consuming is optimal for u_1 while saving is optimal for u_2 under the prior belief. It remains to consider the case when saving is optimal for u_1 and consuming is optimal for u_2 under the prior belief. But in this case, the government can achieve the highest possible payoff by using a non-informative experiment to persuade u_2 to choose consuming with probability one. The discussion is summarized as the following Corollary 1.

Corollary 1. *The government achieves a higher payoff if the consumer becomes less prudent.*

4 Conclusion

This paper studies how the receiver's risk attitudes affect the set of implementable outcomes in a persuasion problem. I consider a two-period consumption model, where the receiver's income in the second period is uncertain and he has the option to transfer his wealth intertemporally through saving. I present a local comparative statics result showing that marginal changes in the receiver's prudence either expand or shrink the set of implementable experiments. Risk aversion does not affect persuasion outcomes, except through pivoting the receiver's optimal action under the prior belief. Moreover, other higher order risk attitudes have no direct impact on persuasion outcomes.

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5 Appendix

For a generic belief π, let F_π be its cumulative distribution function (CDF). With belief π, it is optimal for the receiver to choose saving $\underline{c}$ if and only if

$$E_\pi[U(\underline{c})] \geq E_\pi[U(\overline{c})] \iff \int_0^1 [u(\omega - \underline{c}) - u(\omega - \overline{c}) - u(\overline{c}) + u(\underline{c})] dF_\pi \geq 0 \iff$$

$$\int_0^1 g(\omega) dF_\pi \geq 0, \text{ where } g(z) := \frac{\overbrace{u(z - \underline{c}) - u(z - \overline{c})}^{\text{Change in t = 1 utility}}}{\underbrace{u(\overline{c}) - u(\underline{c})}_{\text{Change in t = 0 utility}}} - 1$$

Define $P(x) := \int_0^x p(z) dF_{\pi_0}(z)$ so $P(1)$ is the *ex ante* probability of receiving the saving signal. His posterior distribution $\pi_{\underline{c}}$ after receiving the saving signal is given by the CDF

$$F_{\pi_{\underline{c}}}(x) = \frac{\int_0^x p(z) dF_{\pi_0}(z)}{\int_0^1 p(z) dF_{\pi_0}(z)} = \frac{P(x)}{P(1)}$$

Similarly, after receiving the signal of consuming, the receiver's posterior CDF is

$$F_{\pi_{\overline{c}}}(x) = \frac{\int_0^x (1 - p(z)) dF_{\pi_0}(z)}{\int_0^1 (1 - p(z)) dF_{\pi_0}(z)} = \frac{F_{\pi_0}(x) - P(x)}{1 - P(1)}$$

The following lemmas are useful for proof of Theorem 1.

Lemma 2. *For any $u \in \mathcal{U}$, the function g is smooth, strictly decreasing and $g(\overline{c} + \underline{c}) = 0$.*

Proof. g is smooth because u is smooth. Since $z - \underline{c} > z - \overline{c}$, $g'(z) = \frac{u'(z-\underline{c}) - u'(z-\overline{c})}{u(\overline{c}) - u(\underline{c})} < 0$ by $u'' < 0$. $g(\overline{c} + \underline{c}) = 0$ comes immediately from the construction. $\square$

Lemma 3. *For continuous functions $m_1, m_2, q_1, q_2 : [0,1] \to \mathbb{R}^{++}$,*

$$\frac{m_1(x)}{q_1(x)} > \frac{m_2(y)}{q_2(y)} \quad \textit{for all } x, y \in [0,1] \implies \frac{\int_0^1 m_1(x) dx}{\int_0^1 q_1(x) dx} > \frac{\int_0^1 m_2(x) dx}{\int_0^1 q_2(x) dx}$$

Proof. I first show

$$\inf_x \frac{m_1(x)}{q_1(x)} \leq \frac{\int_0^1 m_1(x) dx}{\int_0^1 q_1(x) dx} \leq \sup_x \frac{m_1(x)}{q_1(x)} \tag{3}$$

Since $\frac{m_1(x)}{q_1(x)}$ is a continuous function on a closed interval $[0,1]$, it obtains an infimum at

$\underline{x} \in [0,1]$ and a supremum at $\overline{x} \in [0,1]$.

$$\inf_x \frac{m_1(x)}{q_1(x)} := \frac{m_1(\underline{x})}{q_1(\underline{x})} \leq \frac{\int_0^1 m_1(x)dx}{\int_0^1 q_1(x)dx} \iff$$
$$m_1(\underline{x}) \cdot \int_0^1 q_1(x)dx \leq q_1(\underline{x}) \cdot \int_0^1 m_1(x)dx \iff$$
$$\int_0^1 [m_1(\underline{x}) \cdot q_1(x) - q_1(\underline{x}) \cdot m_1(x)]dx \leq 0$$

By the definition of the infimum, $\frac{m_1(\underline{x})}{q_1(\underline{x})} \leq \frac{m_1(x)}{q_1(x)} \iff$ the integrand $m_1(\underline{x}) \cdot q_1(x) - q_1(\underline{x}) \cdot m_1(x) \leq 0$ and hence the first inequality of (3) holds. The proof for the second inequality in (3) is symmetric. For $\frac{m_2}{q_2}$, similar inequalities as (3) hold. Note that

$$\frac{m_1(x)}{q_1(x)} > \frac{m_2(y)}{q_2(y)} \quad \text{for all } x, y \in [0,1] \implies \inf_x \frac{m_1(x)}{q_1(x)} > \sup_y \frac{m_2(y)}{q_2(y)}$$

Since $\frac{m_1}{q_1}, \frac{m_2}{q_2}$ are all continuous functions on a closed interval, the above inequality must be strict. It follows

$$\frac{\int_0^1 m_1(x)dx}{\int_0^1 q_1(x)dx} \geq \inf_x \frac{m_1(x)}{q_1(x)} > \sup_y \frac{m_2(y)}{q_2(y)} \geq \frac{\int_0^1 m_2(x)dx}{\int_0^1 q_2(x)dx}$$

□

5.1 Proof of Proposition 1

Proof.

$$\begin{aligned}(\overline{c}, \omega - \overline{c}) \succeq_1 (\underline{c}, \omega - \underline{c}) &\iff u_1(\overline{c}) + u_1(\omega - \overline{c}) \geq u_1(\underline{c}) + u_1(\omega - \underline{c}) \\ &\iff u_1(\overline{c}) - u_1(\underline{c}) \geq u_1(\omega - \underline{c}) - u_1(\omega - \overline{c}) \\ &\iff \int_{\underline{c}}^{\overline{c}} u_1'(c)\ dc \geq \int_{\omega-\overline{c}}^{\omega-\underline{c}} u_1'(c)\ dc \\ &\iff \omega \geq \overline{c} + \underline{c} \text{ since } u_1'' < 0\end{aligned}$$

Similarly, $(\overline{c}, \omega - \overline{c}) \succeq_2 (\underline{c}, \omega - \underline{c}) \iff \omega \geq \overline{c} + \underline{c}$. □

5.2 Proof of Lemma 1

Proof. In the first case, the constraint (2) is satisfied if and only if

$$\int_0^1 [u(\omega - \underline{c}) - u(\omega - \overline{c}) - u(\overline{c}) + u(\underline{c})] dF_{\pi_0} \leq [u(\overline{c}) - u(\underline{c})] \int_0^1 g(z) \cdot p(z) dF_{\pi_0}$$

Note that the left hand side is equal to $E_{\pi_0}[U(\underline{c})] - E_{\pi_0}[U(\overline{c})]$, which is ≤ 0 by the assumption that the receiver chooses consuming under the prior belief π_0. The right hand side is ≥ 0 because the constraint (1) is satisfied and $E_{\pi_{\underline{c}}}[U(\underline{c})] \geq E_{\pi_{\underline{c}}}[U(\overline{c})] \iff \int_0^1 g(z) \cdot p(z) dF_{\pi_0} \geq 0$. It follows the constraint (2) is redundant. The proof for the second case is symmetric. $\square$

5.3 Proof of Theorem 1

Proof. Define a virtual consumer with $-g(z)$ as its utility function over consumption z. By Lemma 2, $-g(z)$ is smooth, strictly increasing and strictly concave. Denote the Arrow-Pratt measure of absolute risk-aversion of $-g$ as $r(z)$.

$$r(z) := -\frac{-g''(z)}{-g'(z)} = -\frac{g''(z)}{g'(z)} = -\frac{\int_{\underline{c}}^{\overline{c}} u'''(z-x)dx}{\int_{\underline{c}}^{\overline{c}} u''(z-x)dx}$$

Let u_1 be more prudent than u_2 and denote their corresponding g functions as g_1, g_2, respectively. Because u_1 is more prudent than u_2, that is, for all $x, z, \hat{x}, \hat{z}$,

$$-\frac{u_1'''(z-x)}{u_1''(z-x)} > -\frac{u_2'''(\hat{z}-\hat{x})}{u_2''(\hat{z}-\hat{x})}$$

By Lemma 3, for all z,

$$r_1(z) = -\frac{\int_{\underline{c}}^{\overline{c}} u_1'''(z-x)dx}{\int_{\underline{c}}^{\overline{c}} u_1''(z-x)dx} > -\frac{\int_{\underline{c}}^{\overline{c}} u_2'''(z-x)dx}{\int_{\underline{c}}^{\overline{c}} u_2''(z-x)dx} = r_2(z)$$

So the virtual consumer $-g_1$ is more risk averse than the virtual consumer $-g_2$.

To prove the first part of the theorem, suppose the consuming action is optimal for both u_1, u_2 under the prior belief. Suppose p_2^* is an experiment for u_2 that satisfies the obediance constraint (1). By Lemma 1, the obediance constraint (2) is redundant and p_2^* is implementable. To prove p_2^* is also implementable for u_1, it is sufficient to show p_2^* satisfies the u_1's obediance constraint (1).

If $P_2^*(1) = 0$, then $p_2^* = 0$ almost everywhere so p_2^* is a non-informative experiment, which is implementable for u_1. If $P_2^*(1) > 0$, let $H(x) := \frac{P_2^*(x)}{P_2^*(1)}$ be the posterior CDF after receiving the signal of saving. Note that the constraint (1) for u_2 is satisfied if and only if

$$\int_0^1 g_2(z)p_2^*(z)dF_{\pi_0} \geq 0 \iff \frac{1}{P_2^*(1)}\int_0^1 [-g_2(z)]p_2^*(z)dF_{\pi_0} = E_H[-g_2(\tilde{Z})] \leq 0$$

, where $\tilde{Z}$ is a real valued random variable distributed according to H. In other words, the constraint (1) for u_2 is satisfied if and only if the expected utility of $\tilde{Z}$ for the virtual consumer g_2 is non-positive. A similar derivation shows it remains to prove $E_H[-g_1(\tilde{Z})] \leq 0$.

Define the risk premiums t_1, t_2 for $-g_1, -g_2$ under the distribution H as follows:

$$E_H[-g_2(\tilde{Z})] = -g_2(E[\tilde{Z}] - t_2)$$
$$E_H[-g_1(\tilde{Z})] = -g_1(E[\tilde{Z}] - t_1)$$

By Theorem 1 in Pratt (1964), the risk premium t_1 for $-g_1$ must be larger than the risk premium t_2 for $-g_2$. Note that $E_H[-g_2(\tilde{Z})] = -g_2(E[\tilde{Z}] - t_2) \leq 0 \implies E[\tilde{Z}] - t_2 \leq \underline{c} + \overline{c}$ because $-g_2$ is increasing and $-g_2(\underline{c}+\overline{c}) = 0$ by Lemma 2. So $t_1 > t_2 \implies E[\tilde{Z}] - t_1 < \underline{c}+\overline{c}$. It follows $E_H[-g_1(\tilde{Z})] = -g_1(E[\tilde{Z}] - t_1) < 0$ and the experiment p_2^* is also implementable for u_1. The proof for the second case is symmetric. □

5.4 Proof of Proposition 2

Proof. It is optimal for u_1 to choose consuming under the prior belief if and only if

$$0 \leq \int_0^1 -g_1(z)dF_{\pi_0} := E_{\pi_0}(-g_1(\omega))$$

In the proof of Theorem 1, it is shown $-g_1$ must be more risk averse than $-g_2$. It follows $E_{\pi_0}(-g_1(\omega)) \geq 0 \implies E_{\pi_0}(-g_2(\omega)) \geq 0$. So u_2 must also prefer consuming under the prior belief.

To prove the second part, suppose under the prior belief, saving is optimal for u_1 and consuming is optimal for u_2. That is, $\int_0^1 g_1(z)dF_{\pi_0} > 0$ and $\int_0^1 g_2(z)dF_{\pi_0} \leq 0$. Then the experiment p_2 that recommends consuming for all states, $p_2(\omega) = 0$, is not implementable for u_1 because it violates the obedience constraint (2). It can be checked that p_2 satisfies all the obedience constraints for u_2 so it is implementable for u_2. Similarly the experiment p_1 that recommends saving for all states, $p_1(\omega) = 1$, is implementable for u_1 but not for u_2. □